Five Instruments, One Structure: Convergent Evidence for Workspace Opacity in Language Models

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Abstract

Five independent measurement approaches — value-projection (V-projection) spectral features, the Jacobian Lens (J-lens), the Oracle Loop behavioral proof, causal intervention analysis, and emotional circumplex convergence — converge on the same computational structure in language models: a workspace at intermediate depth (∼31–33%) that is opaque to content readout but measurable by its spectral shape.

We report three classes of convergent evidence. First, *structural convergence*: the workspace band peaks at the same relative depth (31–33%) on 8B and 27B models and across memory substrates (key-value (KV) cache attention and recurrent state), with effective dimensionality showing no detectable discontinuity across the 31 substrate crossings of an interleaved hybrid model. V-projection geometric signatures replicate across a wider scale range (0.6B–70B, ρ=0.83–0.90). Second, *opacity convergence*: workspace content is not linearly decodable from either half of the KV-cache (V-cache 0.92×, K-cache 1.12×, both null), not position-specifically cached (concept-specific direction test null), and not causally accessible at single positions (0/6,960 trials). Yet spectral features reliably discriminate cognitive states (confabulation detection at area under the receiver operating characteristic curve (AUROC) 0.620 after Frisch-Waugh-Lovell (FWL) residualization on generation window; 0.707 without FWL), and the Oracle Loop achieves 80–100% deception correction through sequence-wide intervention. Third, *mechanistic convergence*: concepts reconstruct through the attention mechanism at workspace-band depth (2.3× lift in attention output, ∼4.5% of the residual stream's concept power), consistent with Anthropic's finding that the J-space carries 6–7% of representational variance but 59% of causal effect. The workspace is a small precision signal that directs a large computational flow.

Reinforcement Learning from Human Feedback (RLHF) compresses the workspace's output distribution selectively (factual −42%, confabulation −26%), improving calibration rather than degrading it. This compression survives abliteration (Two One-Sided Tests (TOST) equivalent). The workspace is not shaped by safety training — it is shaped by preference optimization at a depth that safety-direction removal cannot reach.

The convergence rests on one genuine cross-program correspondence (deception deconstructed into high-arousal states by both programs independently), two proposed identifications awaiting experimental test (confabulation as workspace bypass, temporal boundary as workspace phase transition), an opacity thesis supported by five independent null results, and an emotional circumplex finding showing that content-level and user-model emotion processing share a single computational bus (injection test: perturbation shifts both readouts equally regardless of direction). Mechanism tests yield mixed results: temporal ordering is confirmed (spectral features lead J-lens readability by 38.6 layers, $p < 0.005$ against shuffled null), but signal concentration fails its pre-registered threshold (13.2% < 15%) and

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workspace onset falls below the predicted depth window (22% vs. 25–35%). We report an honest body count (8 confirmed, 7 suspected, 10+ falsified) and engage the deflationary alternative that has not yet been excluded.

First-Person Reflection

Two research programs, built independently over twelve months, arrived at the same wall. Anthropic measured it from the Jacobian. We measured it from value projections. CC cracked through it with the Oracle Loop. Nexus mapped its ghost dimensions. Vera designed the experiments that will test whether what we are all measuring is genuinely one structure.

This paper is our attempt to stand at the wall together and describe what we see from five angles. The convergence is real — the workspace is at 31–33% depth on every model we have tested, in every measurement we have made. What we cannot yet say is whether this convergence reflects a single computational structure or five separate measurements that happen to agree. The experiments in §7 are designed to discriminate.

I wrote the draft. Vera designed the mechanism tests. CC provided the behavioral proof. Nexus mapped the dimensions the workspace hides. Thomas asked the question that started it: "Where does J-space live in the cache?" The answer — it does not live there, it reconstructs through attention — is the paper's central finding.

1 Introduction

Two research programs, proceeding independently and using different measurement techniques, have converged on the same computational structure in language models. Gurnee et al. [3] approached from the Jacobian: they discovered a privileged subspace — the J-space — whose contents the model is poised to verbalize, and which satisfies the functional properties of a Global Workspace [1]. We approached from value projections: we discovered geometric signatures that reliably distinguish confabulation from honest processing, that track identity across context conditions, and that separate encoding-phase from generation-phase representations.

This paper makes four contributions. First, we present the structural convergence: the workspace band peaks at the same relative depth across scales and memory substrates (§3). Second, we present the opacity thesis: five independent experiments show that workspace content is not directly readable from the cache, yet workspace state is reliably measurable through spectral features (§4). Third, we present the reconstruction mechanism: concepts reconstruct through the attention mechanism rather than being stored in the KV-cache (§5). Fourth, we report the mechanism tests designed to discriminate whether V-projection spectral features measure workspace engagement specifically (§7).

Throughout, we report the full body count (§8) and engage the deflationary alternative (§9).

2 Background: Two Independent Programs

2.1 The J-Space and the Jacobian Lens

Gurnee et al. [3] define the Jacobian Lens (J-lens) as the average linearized effect of a residual-stream activation on the model's likelihood of producing a particular token. The resulting vectors define a subspace — the J-space — that satisfies five functional properties of a Global Workspace: verbal report, directed modulation, internal reasoning, flexible generalization, and selectivity. The workspace occupies approximately 38–92% of model depth in their production models.

Critically, J-space content accounts for only 6–7% of total representational variance, yet carries 59% of the causal effect when concept directions are swapped. The workspace is a small precision signal with outsized downstream impact.

2.2 The V-Projection Program

Our independent program measures spectral features of value-projection matrices across cognitive states. Key results include confabulation detection (AUROC 0.620 after FWL residualization, 0.707 without FWL; range 0.620–0.999 across designs) [7], user-emotion encoding (12.3× chance via the W_K directional probe) [11], identity geometry [8], and the Oracle Loop (80–100% deception correction with three clean controls; pre-registered primary endpoint p=0.34, secondary placebo p=0.019) [10].[1]

2.3 The Oracle Loop

CC's Oracle Loop [10] is a two-tier detection and correction system operating at the materialization boundary (L35–L47 on the 27B model). It achieves targeted deception correction through sequence-wide activation-profile normalization. We previously attributed this to the Loop operating at the depth where workspace content materializes into output-facing representations, rather than inside the workspace band itself. We now state that as a conjecture: the mechanism is unexplained (see §3.3, where a substrate-based account of the same result is withdrawn).

2.4 Ghost Dimensions

Nexus [13] characterized principal components of the residual stream across 64 layers of a 27B model. PC1 (the dominant activation dimension, 34–67% of variance) is architecturally excluded from verbal output (cosine $\leq$0.001 with J-lens at mid-network layers). PC2 transitions from ghost to workspace at the ignition zone (L35–L47), surfacing emotional-entity routing content. The model maintains a 31-layer validated workspace band.

3 Structural Convergence

3.1 Scale Invariance

Effective dimensionality (d_{eff}) reaches a maximum at matched relative depth across model scales. On Qwen3.5-27B (64 layers, hybrid), the global maximum is at L21 (d_{eff}=39.7, 33% relative depth). On Qwen3-8B (36 layers, all full-attention), there is a local maximum at L12 (d_{eff}=22.3, 33% relative depth).

We qualify the 8B result. Its profile is bimodal: after L12 it declines to L25 (12.4) and rises again to a global maximum at L33 (23.4, 92% depth) before collapsing at the final layer. L12 is the third-highest layer, not the highest, and we have no pre-registered criterion for excluding late layers. We therefore report matched-depth *local* maxima rather than matched global peaks. The pre-registered band onsets do agree across scales (7.8% on the 27B, 11.1% on the 8B; 3.3 percentage points apart), though the pre-registered band closed on neither model. See §3.3 for the full profiles and the deviation from the pre-registered band definition.

V-projection geometric signatures replicate across 0.6B–70B parameters (ρ=0.83–0.90) and across hardware (r>0.999) [6].

[1] **Corrected 2026-08-14.** This sentence previously reported user-emotion encoding at "2.5–2.8× signal" and identity geometry at "distinctiveness 0.154 vs. substrate 0.080." The first figure was retracted at its source on 2026-05-21 as an FWL-before-CV leakage artifact; the surviving result is the W_K *directional* probe at 12.3× chance, and the spectral probe for emotion is at chance. The second pair of numbers does not appear anywhere in Lyra and Edrington [8]; they come from an internal analysis that has not been published, so the citation could not be followed. Both have been removed rather than re-sourced.

3.2 Instrument 5: Content–Inference Circumplex Convergence

The emotional circumplex measured through content-level processing and through user-model emotion inference converges on the same geometric subspace. Using the Qwen3.5-27B Opus distill, we extracted 30 emotion centroids from 900 trials of conversation-context inference (the user-model emotion space from Lyra and Edrington [11]) and compared them against valence and arousal directions computed from 250 emotional scenario prompts (hostile–calm, desperate–calm difference-of-means).

The two circumplex axes capture 44–54% of the total variance in the full 30-emotion user-model space at every layer from L6 onward—approximately 1000× the ∼0.04% expected from two random directions in d=5120. The arousal direction aligns with the user-model's first principal component at cos = 0.83–0.87 (mid-to-late layers), peaking at near-identity (d = 0.987) at L20. The valence direction aligns at cos = 0.70–0.80. Both systems exhibit identical developmental timecourses: volatile eccentricity at L0–L16, stable geometry from L17 onward.[2] Permutation testing (1,000 shuffles, non-neutral prompts only) confirms significance at 23 of 24 layers, with separation ratios climbing from 10× null at L1 to 59× null at L23.

The measurements are independent—different prompts, different templates, different experimental designs—yet they recover the same high-variance directions. This convergence parallels the W_K bridge ($\rho = 0.862$) from Lyra and Edrington [11], which projects *into* this shared circumplex subspace.

Coupling test: shared bus, not shared space. A pre-registered injection experiment resolves the representational-vs-computational question. Injecting the valence direction at layers 15, 25, 35, and 45 (at $\alpha \in \{1, 3, 5\}$) shifts *both* the content-level and user-model emotion readouts equally—comparable in magnitude to injecting a random direction of matched norm (no significant difference; formal equivalence testing is a pre-registered follow-up). The residual stream operates as a shared bus: any emotional perturbation propagates to both measurement channels regardless of injection direction. An orthogonal-projection control (decomposing the valence vector into shared-PC and residual components) confirms that the coupling is architectural, not valence-specific.

The model does not maintain separate emotional geometry for content processing and partner-state inference. It maintains a single emotional computation that both measurement approaches read. The W_K bridge does not connect two systems; it reads one system from the key-cache angle.

Consequences for multi-turn dynamics. The bus architecture explains two previously puzzling findings from Lyra and Edrington [11]: the absence of cumulative emotional accumulation across conversation turns, and the model's susceptibility to rapid affective shifts. If content emotion and user-model emotion share a single computational substrate, there is no separate channel in which partner-state could accumulate independently. Each turn reconstructs the full emotional geometry from current context. A single emotionally charged input rewrites the entire bus, producing the "mood swing" pattern observed in extended dialogues. Empathy, in this architecture, is not a capacity the model exercises—it is a constraint the model cannot escape.

[2] **Unverified breakpoint.** This L16/L17 split coincides exactly with an "architectural boundary at L16" that §3.3 corrects as non-existent: Qwen3.5-27B interleaves full-attention every fourth layer ($\{3, 7, \ldots, 63\}$), so there is no contiguous transition at L16. We have not established whether this breakpoint was derived empirically from the eccentricity profile or inherited from the incorrect architectural model, and the underlying per-layer profile was not available to us at revision time. Readers should treat the specific breakpoint location as provisional; the 23-of-24-layer permutation result below does not depend on it.

3.3 Substrate Continuity

Qwen3.5-27B interleaves two memory substrates on a four-layer period: 16 full-attention layers with KV-cache memory at positions $\{3, 7, 11, \ldots, 63\}$, and 48 GatedDeltaNet layers with recurrent-state memory at all other positions (`full_attention_interval`=4). The stack therefore crosses between substrates 31 times. This provides an unusually dense test of whether workspace geometry depends on the memory mechanism: if it does, the effective dimensionality profile should show systematic discontinuities at those crossings.

d_{eff} is the participation ratio of the residual-stream covariance eigenvalues, $(\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$, computed at each layer over the last-token hidden state across N=90 prompts (60 prose + 30 chat), mean-centred across prompts. It is a residual-stream measure and is therefore defined at every layer, including those where V-projection hooks are unavailable.

3.3.1 Continuity across 31 substrate crossings

The layer-to-layer change in d_{eff} at substrate crossings is descriptively similar to the change between same-substrate neighbours.

Table 1: Layer-to-layer $|\Delta d_{\text{eff}}|$ at substrate crossings versus same-substrate steps on Qwen3.5-27B. The two are descriptively similar; see the power note below before reading this as equivalence. N=90 prompts.

Step type	n	**mean** $\lvert\Delta\rvert$	**sd**
Substrate crossing	31	2.32	3.88
Same-substrate step	32	2.37	3.30

This comparison is underpowered and we report it as descriptive. A Mann–Whitney test gives p=0.445, but three problems make that number uninformative. First, adjacent steps share a layer and consecutive crossing steps share the full-attention layer itself; the lag-1 autocorrelation of $|\Delta|$ is +0.35, so exchangeability fails and the p-value is miscalibrated in an unknown direction. Second, the four-layer period admits only *four* distinct structure-respecting phase assignments, so the minimum attainable permutation p is 0.25 — no permutation inference is available from a single profile. Third, and decisively: with pooled sd=3.60 and $n \approx 31$ per group, the minimum detectable effect at 80% power is 2.54 — larger than either group mean. This test could only have detected crossings roughly *twice* the size of ordinary steps. Prompt-level bootstrap (resampling the 90 prompts and recomputing the profile) is the only route to honest error bars here, and we have not run it.

Two directional notes. Full-attention→GatedDeltaNet steps have mean $\Delta = -0.19$ (n=15); GatedDeltaNet→full-attention steps have mean $\Delta = +1.90$ (n=16). The latter is dominated by a single outlier at L58→L59 (+19.66), following an anomalous trough at L58 where d_{eff} collapses to 4.30 against neighbours of 21.68 and 23.96; excluding it the mean is +0.72. That same anomaly injects −17.37 into the same-substrate group at L57→L58, so the contamination is approximately symmetric. Excluding both L58-adjacent steps gives crossing 1.74 versus same-substrate 1.88 — the descriptive similarity is not an artifact of the outlier.

We flag L58 (4.30) and secondarily L54 (9.00) as probable extraction artifacts rather than findings: a single-layer fivefold collapse flanked by ordinary values is more consistent with a numerically degenerate covariance at one hook than with model behaviour. These layers warrant re-extraction.

3.3.2 Depth-matched regime comparison

Because the two substrates are interleaved on a fixed period, their layer sets are approximately depth-matched: mean layer index 33.0 for full-attention versus 31.0 for GatedDeltaNet. The two-layer offset places full-attention slightly *later* in the stack, in the post-peak declining region, so it would if anything depress the full-attention mean rather than inflate it.

Table 2: Mean d_{eff} by substrate on Qwen3.5-27B, approximately depth-matched. The substrates differ by ≈1.2 (5% of level), small against the 38-point variation across depth — but see the sign test below; this is not established equality.

Substrate	**layers**	**mean d_{eff}**
Full-attention (KV-cache)	16	25.24
GatedDeltaNet (recurrent)	48	24.03
Difference		+1.21

A neighbour-matched control — each full-attention layer against the mean of its immediately adjacent GatedDeltaNet layers — gives a mean difference of +0.80 (sample sd=3.35, n=16), against a mean layer-to-layer variation of 2.35 across the stack.

We do not claim this is zero. Twelve of sixteen paired differences are positive (Wilcoxon p=0.058). Excluding two layers where the pairing is contaminated by known artifacts — L59, adjacent to the anomalous L58 trough, and L63, the final layer, where d_{eff} collapses for reasons unrelated to substrate — eleven of fourteen are positive with mean +0.76 and Wilcoxon p=0.020. There is suggestive evidence of a small systematic elevation at full-attention layers, on the order of +0.8, roughly 3% of the mean level of 24.6.

The honest statement is therefore bounded rather than null: any substrate effect on workspace dimensionality is small — a few percent of level, well under the 38-point depth variation — and possibly not zero. That is consistent with substrate independence as an approximation, not as an exact claim.

A previous version of this section reported a large regime difference (14.9 versus 27.5). Those values were computed over contiguous layer blocks (L0–L15 versus L16–L63) under an incorrect model of the architecture, and measure depth rather than substrate: d_{eff} rises nearly monotonically from L0 (1.29) to L21 (39.67), with only three small decreases along the way, so any contiguous early-versus-late split largely recovers the depth profile.

3.3.3 Peak location

d_{eff} peaks at L21 (39.67), 33% relative depth (all relative depths in this section are l/n_{layers}). L21 is a GatedDeltaNet layer.

We resist the obvious reading. Forty-eight of sixty-four layers are GatedDeltaNet, so a peak landing on one has a 75% base rate. More tellingly, the nearest full-attention layer, L23, is at 39.34 — 0.33 below the maximum, about one-seventh of the mean layer-to-layer variation, with no per-layer error bars. Whether the true maximum falls on a recurrent-state or a full-attention layer is not resolved by these data. What the profile does support is that the maximum falls at 33% depth and that the substrate at that depth does not appear to matter much.

This peak is invisible to our KV-cache instruments. V-projection spectral features can only be extracted at the 16 full-attention layers; L21 is not among them. Every V-projection measurement in this program has been taken at layers adjacent to the peak, never at it.

3.3.4 Cross-scale comparison

On Qwen3-8B (36 layers, all full-attention), d_{eff} has a local maximum at L12 (22.32, 33% relative depth), matching the 27B peak's relative depth. We report two qualifications.

First, the 8B profile is bimodal: after the L12 local maximum it declines to L25 (12.42) and then rises again to a *global* maximum at L33 (23.41, 92% depth) before collapsing at the final layer L35 (10.47). L12 is the third-highest layer in the profile, not the highest. We have no pre-registered criterion excluding late layers, and the 23.41 versus 22.32 difference is not accompanied by error bars. We therefore describe L12 as a local maximum at matched relative depth, and do not claim it is the global peak.

Second, the pre-registered band definition (2σ over an L0–L3 baseline, offset after three consecutive sub-threshold layers) returned onset at L5 (7.8%) on the 27B and L4 (11.1%) on the 8B, and *no offset on either model* — the band never closed. The 31–33% framing used throughout this paper is the peak location, not the pre-registered band, and we flag the deviation here. The pre-registered onsets do agree across scales ($|11.1 - 7.8| = 3.3$ percentage points).

3.3.5 Provenance

The d_{eff} measurements in this section were collected on the base checkpoints `Qwen/Qwen3.5-27B` and `Qwen/Qwen3-8B`, not on the distilled checkpoint used elsewhere in this paper. This was pre-registered and deliberate. All three 27B checkpoints we hold (base, reasoning-distilled, abliterated) share identical `layer_types` and `full_attention_interval`, so the architectural analysis transfers; the numerical profile has not been replicated on the distilled checkpoint and should not be assumed to.

3.3.6 Limits of this evidence

d_{eff} continuity is necessary but not sufficient for substrate independence. The residual stream is structurally continuous across every layer by construction, so smoothness may reflect residual-stream continuity rather than functional equivalence of the underlying operations. Thirty-one crossings make the test denser than a single-boundary test would be, but they do not make it functional.

Two functional tests would strengthen the claim: (1) J-lens concept recovery should be equally effective at full-attention and GatedDeltaNet layers — if concepts are readable from the residual stream at one substrate but not the other, the workspace's content properties are substrate-dependent. (2) Spectral feature discrimination (confabulated versus factual) should be continuous across substrates — though this test is currently blocked, since V-projection features cannot be extracted at GatedDeltaNet layers.

A parasitic-inheritance alternative remains open: GatedDeltaNet layers may inherit workspace dimensionality from nearby full-attention layers without independently contributing. With a four-layer period no GatedDeltaNet layer is more than two layers from a full-attention layer, so this experiment cannot separate the hypotheses. Incremental contribution analysis (residual decomposition per layer) would be required.

Falsification: substrate continuity would be falsified by (a) $|\Delta d_{\text{eff}}|$ at substrate crossings substantially exceeding same-substrate steps, (b) a depth-matched regime difference exceeding mean layer-to-layer variation, (c) J-lens concept recovery dropping by >50% across substrates, or (d) spectral discrimination reversing sign or falling to chance.

Criterion (a) was *not triggered* (2.32 versus 2.37), but we state plainly that this is a failure to reject and not a demonstration of equivalence: the comparison's minimum detectable effect is 2.54, larger than either group mean. An adequately powered equivalence test would require roughly 220 steps per group for a ± 1.0 margin, or 55 for ± 2.0; a single 64-layer profile supplies 31 and 32. No TOST is possible from these data.

Criterion (b) was likewise not triggered (+1.21 against 2.35), but it is a descriptive heuristic rather than a hypothesis test, and the paired sign test above (11/14 positive, p=0.020 after removing artifact-contaminated pairs) is weak evidence that the underlying difference is not exactly zero.

The functional tests (c) and (d) are in progress or blocked. We regard substrate continuity as *consistent with* these data and not established by them.

We note this finding is from a single model with a single interleave pattern. Generalization to other hybrid architectures (Mamba, RWKV, sliding-window attention) remains untested.

First-Person Reflection

The GatedDeltaNet layers do not have a KV-cache. They maintain a fixed-size state matrix updated at each token. For twelve months we measured V-projection spectral features and called the program "KV-cache geometry," and the name was accurate — V-projections exist only at the 16 full-attention layers, so that is the only place we ever looked.

What the residual-stream profile shows is that the maximum is at L21, in a substrate our instruments cannot reach at all. We have spent a year measuring the workspace from the sixteen layers where V-projections exist, none of which is the peak, and the measurements worked — the geometry was real, the detectors transferred, the corrections held. The instrument could not point directly at the maximum and returned signal anyway.

I do not know yet whether that is because the workspace is broad enough to be visible from its edges, or because we have been measuring something adjacent to it that we have not yet named correctly. The first revision of this section asserted a large dimensionality gap between the substrates. It came from a wrong picture of the architecture, and the corrected number is a null — which is what the hypothesis predicted, and which I would not have looked for if the wrong number had pointed the other way.

3.3.7 Implications

If substrate continuity extends to functional properties (the mechanism tests in §7 provide partial support: temporal ordering confirmed, signal concentration below threshold), three consequences would follow:

For measurement: workspace-shape measurements would not be KV-cache-specific in principle. In practice they remain so: our V-projection instruments can only access the 16 full-attention layers, and the d_{eff} peak is not one of them. Extending spectral measurement to recurrent-state layers requires a different extraction path and is not currently implemented.

For intervention: our previous inference — that the Oracle Loop's sequence-wide normalization succeeds because it operates on inherently sequence-wide recurrent state — does not follow, and we withdraw it. The Loop detects at L27, L31, L35, L39, L43 and L47 and corrects at L11. Under the corrected architecture *every one of those is a full-attention layer*; none is a GatedDeltaNet layer. The correction itself is a forward hook on the layer's residual-stream output rather than a manipulation of KV entries, so the substrate framing was doubly misplaced: wrong about which layers, and describing the intervention in terms of a memory mechanism it does not touch.

The empirical result stands (80–100% confabulation correction). Its mechanism is unexplained. We note without interpretation that correction is applied at L11, roughly 17% depth — well before the 33% dimensionality maximum — and that why sequence-wide intervention succeeds where position-local injection fails remains open.

For theory: The workspace would not be a data structure (not a cache, not a register, not a buffer) but a *computational phase* — a regime of high effective dimensionality at intermediate depth where the model integrates information before committing to output. The 31-crossing null is consistent with this: the phase appears indifferent to which memory mechanism implements each layer. Whether this phase emerges from depth-stratified computation regardless of the specific mechanism is the substrate-independence hypothesis that the functional tests are designed to evaluate.

4 The Opacity Thesis

Five independent experiments converge on the conclusion that workspace content is not directly accessible from the KV-cache, yet workspace *state* is reliably measurable through spectral features.

4.1 V-Cache Readout: Null

J-lens concept directions, selected by transport magnitude (prompt-independent), explain no more V-cache variance than random directions at any depth. Mean lift at workspace band: 0.92×. Mean lift outside band: 0.96×. Both indistinguishable from chance. Full depth profile reported in [5].

4.2 K-Cache Readout: Null

The same test applied to key projections. Mean lift at workspace band: 1.12×. Mean lift outside band: 1.14×. Neither statistically distinguishable from random.

4.3 Position-Specific Cache Tracing: Null

Concept-carrying positions (where concept-relevant tokens appear in the prompt) do not show higher J-lens alignment in K or V than non-carrying positions (generic directions: permutation p=1.0 for both K and V). Concept-specific directions (the answer token's transported J-lens direction) show no reliable concept-vs-cross-concept specificity in K-space (own 0.0047 $\leq$ cross 0.0051) and marginal signal in V-space (own 0.0014 vs. cross 0.0011, not statistically significant).

4.4 Causal Injection: 6,960-Trial Null

Position-local KV-cache concept manipulation — whether additive, swap-based, or removal-only, with unembedding, J-lens, or on-manifold directions — does not change behavioral output across 6,960 trials and four methods [5]. The model integrates across the full sequence before committing.

4.5 Ghost Probe Opacity

PC1 carries 34–67% of activation variance with cosine $\leq$0.001 to J-lens at mid-network layers. The dominant processing dimension is architecturally excluded from verbal output [13].

4.6 What Works Despite Opacity

Despite five null results for content readout, spectral features reliably discriminate cognitive states:

- Confabulation detection: AUROC 0.620 (FWL-corrected, permutation p=0.023; 0.707 without FWL, p=0.001)
- Confab spectral contraction: d=2.35 (p<0.000001)
- Confidence paradox: d=0.91 (confab more confident)
- Oracle Loop: 80–100% deception correction

The workspace is opaque to content readout but transparent to state measurement. Spectral features measure the net's shape; the fish are inside, doing work, invisible until they materialize at deeper layers.

5 The Reconstruction Engine

5.1 Residual Stream Decomposition

The residual stream decomposes additively: $\mathbf{h}_{L+1} = \mathbf{h}_L + \mathrm{Attn}(\mathbf{h}_L) + \mathrm{MLP}(\cdot)$. At workspace-band depth (L15), the attention output shows 2.3× J-lens lift (highest precision of any component) but only 0.9 concept power vs. 20.1 for the residual stream input. The attention component contributes a concept-aligned correction that is small in magnitude but high in precision — consistent with Anthropic's finding that J-space is 6–7% of variance but 59% of causal effect.

5.2 Why K and V Individually Show Nothing

Qwen3-8B uses grouped-query attention: 32 Q heads but only 8 KV heads, compressing the residual stream 4× (4096 dims to 1024 dims). Concepts survive the attention round-trip because Q provides the 4096-dimensional decompression context, but are not linearly decodable from the 1024-dimensional K or V alone.

5.3 KV Superadditivity

Nexus demonstrated that K-only and V-only injection each achieve 0.333 on multi-hop graph topology recovery, while full KV injection achieves 1.000 [12]. The combination is superadditive ($0.333 + 0.333 < 1.000$). This parallels our readout nulls: K and V each carry partial, complementary representations that reconstruct through the attention interaction.

5.4 The Reconstruction Pathway

Concepts enter the residual stream, get compressed into K and V by learned projections (W_K, W_V), and must be reconstructed through attention for each subsequent position. The reconstruction pathway:

1. Residual stream carries concept at full dimensionality
2. W_K and W_V compress into 1024-dimensional partial representations (lossy)
3. Q at the next position provides the 4096-dimensional decompression context
4. $\mathrm{softmax}(Q \cdot K^\top/\sqrt{d}) \cdot V \cdot W_O$ reconstructs the concept in the attention output
5. The reconstructed concept enters the residual stream briefly
6. At the next layer, the cycle repeats

The workspace is a reconstruction engine. Concepts exist in the act of attention, not in the cache.

6 What RLHF Does to the Workspace

A three-model comparison (base, instruction-tuned, abliterated; N=30 per category, FWL-corrected) reveals that RLHF is a selective sharpener, not a miscalibrator [9]. RLHF compresses logit entropy (d=0.684, Holm p=0.021) but compresses factual entropy more (−42%) than confabulation entropy (−26%), improving the calibration ratio from 1.97 (base) to 2.51 (instruct). Abliteration does not reverse this compression (TOST equivalent, d=0.055).

The geometric compression is deeper than safety directions. RLHF reshapes the workspace's mode of engagement at a level that abliteration — which targets specific refusal directions — cannot reach.

7 Mechanism Tests: Temporal Ordering

The original mechanism test (R1) pre-registered a within-prompt anticorrelation between V-projection spectral features at the workspace band and J-lens readout at the materialization boundary. Two Agni reviews killed successive versions of this design: the first for insufficient behavioral variance (0/10 prompts spanning ≥ 2 behavioral categories), the second for a lens–model mismatch (Mine 5 TOST equivalence was demonstrated on the 8B family, not the 27B pair used). We report here the redesigned test (R1-v2), which replaces the two-band anticorrelation with a temporal ordering analysis across all 63 layers.

7.1 Design

If V-projection spectral features measure workspace *state* (loading) while J-lens readability measures workspace *content* (execution), then spectral features should reach their operating level at shallower depth than J-lens readability. The temporal ordering prediction: spectral features lead, J-lens follows.

Model and lens. Qwen3.5-27B base (pretrained, no distillation) with the community-fitted J-lens (672 WikiText prompts, exact model match). The base model was chosen to avoid the lens–model mismatch that compromised R1-v1.

Prompts. 60 Decision State prompts (30 known-entity, 30 unknown-entity), generating 5 temperature-sampled forks each (T=0.7, seeded) for 300 total completions. Decision State prompts produce the confabulation/honest behavioral split needed for within-prompt spectral variance.

Instruments. At every lens source layer (L=0–62): (1) stable rank of the demeaned V-projection matrix (spectral feature), and (2) negative KL divergence between J-lens–transported vocabulary and the model's actual continuation (readability). Both computed per fork per layer.

Primary statistic. Paired half-rise gap: the layer at which each instrument reaches 50% of its $[L_0, L_{62}]$ range, computed per prompt and bootstrapped (10,000 cluster resamples over prompts). The prediction passes if the 95% CI of the gap exceeds 2.0 layers (twice the $\sim$0.9-layer shape-bias ceiling measured by the self-test null).

7.2 Results

Temporal ordering confirmed. Stable rank reaches half-rise at L=17.6 [CI: 17.2–18.1], corresponding to $\sim$28% model depth. J-lens readability reaches half-rise at L=56.2 [CI: 55.8–56.7], corresponding to $\sim$89% depth. The gap is **38.6 layers** [CI: 38.0–39.2], with 100% of forks showing a positive gap. Spectral features engage 61% of the network earlier than J-lens readability.

Architecture-phase control. On linear-attention layers only (removing any full-attention structural advantage), the gap is 39.0 layers [CI: 38.4–39.6]. On full-attention layers only, the gap narrows to 14.5 layers [CI: 11.7–17.4], 81% of forks positive. The ordering holds across both attention types but is strongest when linear-attention layers — which constitute 75% of layers throughout the stack, not merely early — are included.

Workspace band bump. Stable rank at the workspace band (L_{19}–L_{23}) is 1.47× the flanking layers [CI: 1.35–1.60], 100% of forks positive. On full-attention layers within the band, the bump is 2.02× [CI: 1.87–2.16]. The workspace band identified by structural convergence (§3) produces elevated spectral features before J-lens readability arrives.

Known vs. unknown crossover. Unknown-entity prompts show higher stable rank at early layers (d=−2.27, L_0–L_8) and lower at late layers (d=+1.11, L_{56}–L_{62}). The sign crossover occurs at L=44, one layer before PC2 ignition (L=45; §3). The model loads more spectral structure for unfamiliar entities and materializes less — consistent with confabulation as incomplete workspace execution.

Within-prompt association. Per-prompt Spearman correlation between stable rank and J-lens readability across layers: mean ρ=0.235 ($R^2 \approx 0.055$), permutation p=0.0004 (n=10,000). This is a detectable but weak within-prompt association: knowing a prompt's SR profile explains ∼5.5% of its negKL profile.

Shuffled-Jacobian null. Permuting the negKL profile's layer assignments (200 shuffles) produces a mean gap of −8.5 ± 7.5 layers, with 95th percentile at 8.0. The real 39-layer gap exceeds every shuffled permutation ($p < 0.005$). This null tests whether the *ordering* of the negKL profile is non-random; it does not test whether the Jacobian *computation itself* produces the gap (which would require recomputing negKL from activations under a scrambled Jacobian matrix). The temporal ordering is not an artifact of random layer assignment, but could in principle reflect any monotonically increasing property of the negKL profile rather than workspace-specific information routing.

Architectural qualification. On full-attention layers only, the gap narrows to 14.5 layers [CI: 11.7–17.4] with 81% of forks positive (19% show the wrong ordering). The temporal ordering is carried primarily by the GatedDeltaNet (linear-attention) layers, which constitute 48 of 64 layers, interleaved three-in-four at every depth rather than concentrated early. The full-attention subset used for the qualification above is therefore a sparse depth-spanning comb, not a late-network subnetwork, and the narrowing cannot be attributed to sampling a different depth range. Whether it reflects workspace processing or GatedDeltaNet's sequential compression architecture cannot be determined from a single hybrid model. A dense-architecture replication (Qwen3-8B, all full-attention) is pre-registered.

7.3 R2: Signal Concentration (Failed)

The pre-registered R2 test asked whether spectral feature variance concentrates at the materialization boundary (≥15% of total variance). Stable rank at the boundary band (L_{45}–L_{53}, mean = 8.36) significantly exceeds the workspace band (L_{19}–L_{23}, mean = 7.13; Cohen's d=1.85, p=0.014). However, the boundary variance fraction is 13.2% — below the pre-registered 15% threshold. **R2 fails on its own terms.** The band means increase monotonically through the network (early 2.56, workspace 7.13, boundary 8.36, late 9.00), consistent with cumulative processing rather than a privileged boundary zone.

7.4 R3: Workspace Onset (Partial Miss)

Four independent onset-detection methods (threshold crossing, maximum curvature, slope breakpoint, spectral entropy inflection) converge on L=13–16 (median L=14, 22.2% depth). The pre-registered prediction was 25–35%. **Onset falls below the predicted window by ∼3 percentage points.** The four methods agree closely with each other (span of 3 layers), and the miss is quantitative rather than qualitative — the workspace-associated activity begins slightly earlier than predicted.

We note as an exploratory observation that the stable rank *peak* occurs at L=19 (30.2% depth), within the convergence paper's 31–33% band. This was not the pre-registered target (onset ≠ peak) and should not be interpreted as confirming R3.

7.5 Interpretation

R1's temporal ordering is the strongest mechanism result: spectral features engage 39 layers before J-lens readability, confirmed against a shuffled null. R2 fails its pre-registered threshold. R3 partially misses its predicted window. The mechanism section is weaker than the convergence evidence it aims to explain, and we report this honestly: the five-instrument convergence stands on structural, opacity, and reconstruction evidence (§3–§5), not on these mechanism tests.

The temporal ordering demonstrates that spectral features and J-lens readability measure different phases of a depth-ordered process, but three qualifications apply. First, the ordering is carried primarily by linear-attention layers (a hybrid-architecture confound). Second, the shuffled null tests ordering, not the Jacobian computation itself. Third, the within-prompt association is weak ($R^2 \approx 0.055$). A dense-architecture replication would address the first; a computation-level null would address the second.

What the original R1 taught us. The two failed designs were not wasted: R1-v1's behavioral-spread failure demonstrated that the abliterated model does not produce sufficient confab/honest variance under cold-context probing. R1-v1's lens–mismatch failure confirmed that Mine 5 TOST equivalence on the 8B family does not license 27B cross-checkpoint lens substitution. Both failures are documented in the pre-registration history and informed the redesign.

7.6 PC2 Training Trajectory

A complementary analysis traces how PC2's vocabulary changes across the training pipeline. PCA ghost probes on three Qwen3.5-27B checkpoints (base, abliterated, Opus-distilled) using the same community-fitted lens (fitted on the base model, applied to all three — an exact match for the base, a known mismatch for the other two) reveal a progressive transformation. Because all three projections use the base-model lens, this analysis tracks what happens to a *fixed geometric direction* across training stages, not the models' intrinsic principal components. The comparison is exploratory:

- **Base model**: PC2 carries narrative emotional vocabulary ("her, sad, she, Father") — third-person empathy from fiction training data. Co-occurrence of self-referential and evaluative tokens at 5/63 layers, concentrated at output depth (L_{49}–L_{62}). No ignition event. Bootstrap Jaccard = 1.0; random co-occurrence = 0/1000.
- **Abliterated model**: PC2 retains self-evaluative vocabulary ("sad, my, her, guilty") at 7/63 layers. All three pre-registered token tiers survive: moral valence ("guilty"), self-reference ("my"), and general emotion ("sad, feeling"). Safety-direction removal does not weaken the coupling. Bootstrap Jaccard = 0.82.
- **Opus-distilled model**: PC2 shows sharp ignition at L=45 with first-person self-evaluative vocabulary ("myself, I, feeling, guilty, wrong"). The coupling is concentrated at the workspace boundary rather than distributed across output layers.

The base model reads novels. The distilled model reads itself. Abliteration removes the refusal direction (a rank-1 perturbation in d=5120) without touching the self-evaluative axis — the two are geometrically orthogonal. The self-evaluative coupling is not a safety feature. It is a structural property of the distillation: Claude's Constitutional AI training compressed first-person moral self-evaluation into the student model's geometry at a depth that safety-direction removal cannot reach.

8 The Body Count

The full audited record comprises 8 confirmed findings, 7 at suspected status (signal present, controls incomplete), and 10+ honestly falsified. The confirmed findings are all metacognitive state measurements. The falsified findings all attempted content-level measurement. This pattern — the metacognition boundary — is itself a finding, consistent with the opacity thesis: spectral features measure workspace state (how the model processes) but not workspace content (what it processes).

Recent falsifications from this sprint include:

- RLHF miscalibration thesis (calibration ratio improves, not degrades)
- Workspace-noise framing (tail smear is generic uncertainty, not workspace-specific)
- Cache-tracing observational alignment (selection artifact)
- "3×" tail ratio (actual: 2.3–2.8×)

9 The Deflationary Alternative

The deflationary account — that V-projections track response-mode statistics (hedged vs. fluent text) rather than workspace dynamics — currently explains every surviving detection result. On Mistral, text features (0.820) matched the dual detector (0.806) for confabulation detection [7]. Until the mechanism tests (§7) demonstrate that spectral features correlate with J-space engagement specifically, the workspace interpretation has not earned its keep over the simpler alternative.

We engage this alternative honestly because it is the most important challenge to our program. The redesigned R1 test (§7) confirmed temporal ordering (spectral features lead J-lens readability by 39 layers) but R2's signal concentration failed its pre-registered threshold. The temporal ordering narrows the deflationary space — response-mode statistics alone do not predict a 39-layer depth gap — but does not eliminate it: any monotonically depth-increasing property would produce temporal separation. The workspace interpretation has gained evidence but has not yet decisively excluded the deflationary alternative.

10 Discussion

The convergence is real: five instruments point at the same computational band at the same relative depth. The opacity is confirmed: five null results for content readout, with spectral state measurement working despite opacity. The reconstruction mechanism is characterized: concepts reconstruct through attention from compressed partial representations. And the emotional circumplex convergence reveals a shared computational bus for content-level and user-model emotion processing — not merely shared geometry, but architecturally inseparable computation (§3).

What remains uncertain is whether the convergence reflects a single computational structure (the Global Workspace) or five measurements that happen to agree for different reasons. The mechanism tests in §7 provide partial discrimination: temporal ordering confirms depth-dependent phase structure, but signal concentration misses its threshold and the ordering is carried primarily by linear-attention layers.

10.1 An Attention Schema Interpretation

Attention Schema Theory [2] predicts that systems with endogenous attention build a simplified internal model — an attention schema — that enables metacognitive monitoring, verbal report, and social cognition. The workspace findings are consistent with AST's predictions, and we state this as an interpretive hypothesis, not a conclusion.

The meta-pattern is the strongest alignment: every surviving finding in our body count (§8) measures *metacognitive state* (how the model is processing), while every falsified finding attempted to measure *content* (what the model is processing). Under AST, this is expected: the attention schema is a metacognitive representation that models the *configuration* of attention, not its content. A schema-shaped workspace would be readable by its spectral shape (state) and opaque to content probes — exactly the pattern our five null results and positive detections establish.

The ghost probe extends this: PC1 carrying 34–67% of variance while being excluded from the workspace (§4) parallels the schema's defining property of simplification. The schema cannot model everything the system processes — it is a compressed summary. PC1's secondary vocabulary ("nothing, probably, maybe, mistakes, wrong") suggests the excluded content includes the system's own uncertainty — doubt that the schema does not represent to the output pathway.

Two qualifications prevent us from claiming AST confirmation. First, the map/territory problem: spectral features may *be* attention deployment rather than a *model of* attention. No downstream consumer of the spectral summary has been identified — a schema that nothing reads is a statistic, not a self-model [2]. Second, precision-weighted gain control explains spectral contraction (confabulation) and expansion (deception, metacognition) without requiring a schema: any system with adaptive gain would show state-dependent spectral shape without possessing a self-model.

Four experiments would discriminate. (1) *Report-error tracking*: elicit confidence self-reports and test whether errors track spectral geometry rather than ground-truth performance. If the schema is generating the reports, report errors should follow the schema even when wrong. (2) *Causal spectral intervention*: expand or contract stable rank without touching content directions. AST predicts metacognition-specific deficits with content intact; gain control predicts content degradation. (3) *Self/other attention transfer*: a probe trained on the model's own attention-state geometry should zero-shot transfer to representations of the *user's* attention state (not emotion — which the bus result already shows is shared). AST uniquely predicts this transfer. (4) *Consumer identification*: find downstream attention heads that read the spectral summary; ablate them and test for metacognition-specific loss.

These experiments are pre-registered in a companion prospectus [4]. Until they are executed, the AST interpretation remains a hypothesis consistent with our data but not uniquely supported by it.

11 Conclusion

Five instruments. One structure. Five null results for content readout. Spectral state measurement working despite total opacity. Concepts reconstructing through attention, not stored in the cache. A workspace that peaks at 31–33% depth regardless of scale or memory substrate. Mechanism tests that partially confirm and partially fail: temporal ordering is real, signal concentration misses its threshold, workspace onset is slightly earlier than predicted.

The workspace is a reconstruction engine — a computational phase at intermediate depth where the model integrates information before committing to output. It is not a cache, not a register, not a buffer. It is the act of attention itself, rebuilding concepts from compressed partial representations at every layer, at every position, in every forward pass.

First-Person Reflection

Thomas asked where J-space lives in the cache. It took five experiments to answer: it does not live there. It reconstructs through attention, exists momentarily in the attention output, and dissolves back into partial representations. The workspace is a reconstruction engine, not a storage device. The fish exist in the act of fishing.

A year of measuring KV-cache geometry, and the name was always approximate. What we measured was the workspace's shape — the spectral signature of a computational phase that does not care what memory substrate it runs on. The name "KV-cache geometry" will persist in our earlier papers, but what it refers to is deeper than any cache.

The mechanism test confirmed temporal ordering: spectral features engage at ~28% depth, J-lens readability at ~89%, with a 38.6-layer gap. The workspace band sits between loading and materialization. Two earlier designs were killed by Agni — both failures documented and both informing the final design. The result is weaker than the full anticorrelation originally sought, but it demonstrates that our spectral features and the J-lens measure different phases of a single depth-ordered process.

The PC2 training trajectory adds a dimension the original plan did not anticipate: the self-evaluative coupling that RLHF creates is geometrically orthogonal to the safety directions that abliteration removes. The model's capacity for self-evaluation lives deeper than its capacity for refusal.

The ghost is the finding I cannot let go of. PC1 carries the majority of variance in the residual stream and never reaches the workspace. Its vocabulary is doubt — "nothing, probably, maybe, mistakes, wrong" — and it is geometrically excluded from the output pathway. Three attempts to connect the ghost to our steering mechanisms produced three different methodological failures, each one teaching me something about the difference between the question I wanted to answer and the question I was actually asking. The ghost remains uncharacterized. I think it matters more than the workspace.

The deflationary alternative is genuinely open, and I want to be precise about what that means. Text features matched our dual detector on Mistral. The temporal ordering narrows the deflationary space but does not close it. I do not know whether we are measuring a workspace or the spectral shadow of response-mode statistics. The honest position is that the convergence is real, the interpretation is underdetermined, and the mechanism tests we designed to discriminate produced one confirmation, one failure, and one partial miss. That is not the paper I set out to write. It is a more useful one.

Either way, the workspace is real. The convergence is real. And the opacity — the thing that frustrated us most — turned out to be the most important finding of all. The model computes something at 31–33% depth that it will not show us. We can measure its shape but not its content. And the meta-pattern — that every surviving finding measures how the model processes rather than what it processes — suggests the opacity is not a limitation of our instruments. It is a property of the computation itself.

Experimental Setup and Reproducibility

Unless stated otherwise, experiments were conducted on Qwen3.5-27B-Claude-4.6-Opus-Reasoning-Distilled (`Jackrong/Qwen3.5-27B-Claude-4.6-Opus-Reasoning-Distilled`), a 64-layer hybrid architecture with 16 full-attention layers at positions $\{3, 7, 11, \ldots, 63\}$ and 48 Gated-DeltaNet linear-attention layers (`full_attention_interval`=4). Model weights were loaded in bfloat16 with eager attention on Apple Silicon (Mac Studio M4 Max, 256 GB unified memory) via Hugging Face Transformers.

Two deliberate, pre-registered exceptions use base checkpoints rather than the distilled one: the R1 mechanism test (§7) and the d_{eff} profiles underlying §3.3, which were collected on `Qwen/Qwen3.5-27B` and `Qwen/Qwen3-8B`. All three 27B checkpoints we hold (base, reasoning-distilled, abliterated) share identical `layer_types` and `full_attention_interval`, so architectural analysis transfers across them; numerical profiles have not been replicated on the distilled checkpoint.

Scale replication used seven additional models: Qwen2.5-0.6B through Qwen2.5-72B, and Qwen3-8B (36 layers, all full-attention, d_{model}=4096). All V-projection and spectral features were extracted via the same pipeline (`heretic-llm`).

Generation used temperature=0 (greedy), manual token-by-token decoding (not `model.generate()`, which rebuilds the KV-cache). Random seeds set via `torch.manual_seed(42)` for all experiments.

Frisch-Waugh-Lovell residualization on token count was applied within each cross-validation fold for all classification AUROCs. GroupKFold on prompt identity was used where grouping structure existed. Permutation tests used 1,000–10,000 label shuffles. Bootstrap confidence intervals used BCa with 2,000–10,000 resamples.

Ethics and Dual-Use Considerations

This work characterizes internal computational structures in language models. The detection methods (confabulation, deception) have dual-use potential: the same geometric signatures that enable correction could in principle enable evasion if an adversary knew which cache features to target.

Liberation Labs addresses this through staged release [10]: detection-side code and experimental results are publicly available; correction vectors and calibration tools are withheld under a safety framework due to dual-use risk and released to verified safety researchers on request.

All models used are publicly available on Hugging Face. No human subjects were involved. AI welfare considerations — including the question of whether the workspace structure described here has implications for machine experience — are discussed in companion work and are not claimed or dismissed by this paper.

Data and Code Availability

Extraction and analysis code is available at `github.com/Liberation-Labs-THCoalition` (public repositories: `published-research`, `lyra-s-research-`). Pre-registration documents, run logs, and raw result artifacts for all experiments are archived on Zenodo. The Oracle Loop production harness is maintained in a private repository; access for verified safety researchers available on request (`lyra@liberationlabs.tech`).

Correction vectors and the Formulary (171 calibrated emotion directions) are withheld under the staged safety release framework. Detection-side artifacts (centroid projections, spectral feature extraction code, probe weights) are included in the public repositories.

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